

Physics informed neural network of a moving conductive string in a magnetic field

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1 Physical Model

Consider an infinite conductive string in a homogeneous magnetic field, $\mathbf{B}_0 = B_0 \mathbf{i}_x$, where B_0 is a constant. The string is being pulled at a constant velocity, $\mathbf{V} = V_0 \mathbf{i}_x$, through two rings which are located a distance l from each other. The motion of the string is considered in a fixed Cartesian coordinate system (x, y, z) , with unit vectors $\mathbf{i}_x, \mathbf{i}_y, \mathbf{i}_z$. The center of one of the rings is assumed to be the center of the coordinate system.

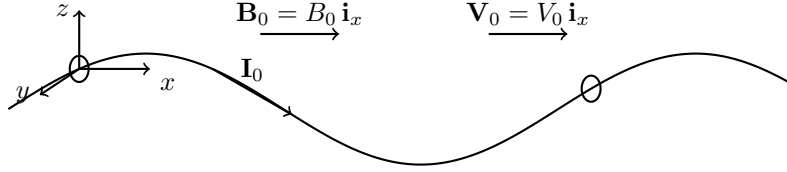


Figure 1: Schematic of the string with the applied fields and forces.

The influence of gravity and other external forces are neglected. The unit length and the position of each material point of the string is assumed to be constant. We identify the position of each material point of the string through its arc length $s \in [0, l]$. In Euclidean space, we denote the displacement of the string by $\mathbf{r}(s, t) = x(s, t)\mathbf{i}_x + y(s, t)\mathbf{i}_y + z(s, t)\mathbf{i}_z$. We also assume that $\mathbf{k} = \frac{\partial x}{\partial s}\mathbf{i}_x + \frac{\partial y}{\partial s}\mathbf{i}_y + \frac{\partial z}{\partial s}\mathbf{i}_z$ represents the unit vector directed along the tangent to the deformed line of the string. If we assume a current $\mathbf{I} = I_0(t)\mathbf{k}$ flows through the string, then the magnetic field will exert forces on the string. In this particular case, the forces acting on the string are called Laplace forces given by

$$\mathbf{F} = I_0 B_0 (\mathbf{k} \times \mathbf{i}_x). \quad (1)$$

The motion of the string is described by

$$\frac{\partial \mathbf{Q}}{\partial s} + \mathbf{F} = \rho \mathbf{R}, \quad \mathbf{Q} \times \mathbf{k} = 0, \quad \left(\frac{\partial \mathbf{r}}{\partial s}\right) = 1, \quad (2)$$

where $\mathbf{Q}$ is the internal tension in the string, ρ is the mass density per unit length, and $\mathbf{R}$ is the inertial force of the string given by

$$R_x = 0, \quad R_y = \frac{\partial^2 y}{\partial t^2} + 2V_0 \frac{\partial^2 y}{\partial s \partial t} + V_0^2 \frac{\partial^2 y}{\partial s^2} + \eta \frac{\partial y}{\partial t}, \quad R_z = \frac{\partial^2 z}{\partial t^2} + 2V_0 \frac{\partial^2 z}{\partial s \partial t} + V_0^2 \frac{\partial^2 z}{\partial s^2} + \eta \frac{\partial z}{\partial t}, \quad (3)$$

where η is a damping coefficient. The inertial force in the x-direction is zero since we assume that the string is being pulled in the x-direction with a constant velocity. The complex function

$u(s, t) = y(s, t) + iz(s, t)$ will be used to represent the displacement of the string in the y and z directions.

Using the aforementioned equations, we can obtain the governing equations for lateral vibrations

$$\rho \left[u_{tt} + \eta u_t + 2V_0 u_{st} + V_0^2 u_{ss} \right] = \frac{\partial}{\partial s} \left(\frac{P_0 u_s}{\sqrt{1 - |u_s|^2}} \right) - iI_0 B_0 u_s, \quad (4)$$

where $0 < s < l$ and $P_0 = Q_x$ is the tension applied to the string at rest. The boundary conditions are given by

$$u(0, t) = 0, \quad u(l, t) = 0, \quad t \geq 0. \quad (5)$$

These boundary conditions specify that the string is fixed at the rings. Initial conditions will be assigned appropriately. if we assume that the deformations are negligibly small, i.e., $|u_s| \ll 1$, then the governing equation can be linearized as

$$\frac{\rho}{P_0} \left[u_{tt} + \eta u_t + 2V_0 u_{st} \right] = \left(1 - \frac{\rho V_0^2}{P_0} \right) u_{ss} - i \frac{I_0 B_0}{P_0} u_s, \quad (6)$$

where $0 < s < l$ and $t \geq 0$.

2 PINN Formulation